

UNIVERSITY NAME

*Analysis IV — Final Exam**Year: 2019***Exercise 1:**

Determine the nature (convergence/divergence) of the following sequences:

$$\begin{aligned} \text{a) } u_n &= \cos \left(\pi \frac{2n^3 + n^2 + an + b}{2n^2} \right), \quad a, b \in \mathbb{R} \\ \text{b) } u_n &= \left(1 - \frac{1}{n^a} \right)^n, \quad a > 0 \end{aligned}$$

Answer Area

a) For $u_n = \cos \left(\pi \frac{2n^3 + n^2 + an + b}{2n^2} \right)$:

Simplify the fraction:

$$\frac{2n^3 + n^2 + an + b}{2n^2} = n + \frac{1}{2} + \frac{a}{2n} + \frac{b}{2n^2}$$

The argument becomes:

$$\pi \left(n + \frac{1}{2} + \frac{a}{2n} + \frac{b}{2n^2} \right) = \pi n + \frac{\pi}{2} + \frac{a\pi}{2n} + \frac{b\pi}{2n^2}$$

Since $\cos(\theta + \pi n) = (-1)^n \cos \theta$, we have:

$$u_n = \cos \left(\pi n + \frac{\pi}{2} + \frac{a\pi}{2n} + \frac{b\pi}{2n^2} \right) = (-1)^n \cos \left(\frac{\pi}{2} + \frac{a\pi}{2n} + \frac{b\pi}{2n^2} \right)$$

Using $\cos \left(\frac{\pi}{2} + \epsilon \right) = -\sin \epsilon$, we get:

$$u_n = (-1)^n \left[-\sin \left(\frac{a\pi}{2n} + \frac{b\pi}{2n^2} \right) \right] = (-1)^{n+1} \sin \left(\frac{a\pi}{2n} + \frac{b\pi}{2n^2} \right)$$

As $n \rightarrow \infty$, $\frac{a\pi}{2n} + \frac{b\pi}{2n^2} \rightarrow 0$, so $\sin \left(\frac{a\pi}{2n} + \frac{b\pi}{2n^2} \right) \sim \frac{a\pi}{2n}$.

Therefore:

$$u_n \sim (-1)^{n+1} \frac{a\pi}{2n}$$

This shows that:

- If $a \neq 0$, u_n converges to 0 (alternating but decreasing in absolute value).

- If $a = 0$, then $u_n \sim (-1)^{n+1} \frac{b\pi}{2n^2}$ converges to 0 faster.
- In all cases, $\lim_{n \rightarrow \infty} u_n = 0$.

b) For $u_n = \left(1 - \frac{1}{n^a}\right)^n$ with $a > 0$:

Take the natural logarithm:

$$\ln u_n = n \ln \left(1 - \frac{1}{n^a}\right)$$

Using the asymptotic expansion $\ln(1+x) = x - \frac{x^2}{2} + o(x^2)$ as $x \rightarrow 0$:

$$\ln \left(1 - \frac{1}{n^a}\right) = -\frac{1}{n^a} - \frac{1}{2n^{2a}} + o\left(\frac{1}{n^{2a}}\right)$$

So:

$$\ln u_n = -\frac{n}{n^a} - \frac{n}{2n^{2a}} + o\left(\frac{n}{n^{2a}}\right) = -n^{1-a} - \frac{1}{2}n^{1-2a} + o(n^{1-2a})$$

Now analyze based on a :

- If $0 < a < 1$: Then $1 - a > 0$, so $\ln u_n \sim -n^{1-a} \rightarrow -\infty$, thus $u_n \rightarrow 0$.
- If $a = 1$: Then $\ln u_n = -1 - \frac{1}{2n} + o(1/n) \rightarrow -1$, so $u_n \rightarrow e^{-1}$.
- If $a > 1$: Then $1 - a < 0$, so $n^{1-a} \rightarrow 0$, and $\ln u_n \rightarrow 0$, thus $u_n \rightarrow 1$.

Exercise 2:

Consider the series of functions $\sum_{n \geq 1} f_n$ where

$$f_n(x) = \frac{x}{n(1+n^2x)}.$$

Let $f(x) = \sum_{n=1}^{\infty} f_n(x)$ when it exists.

1. Show that f is defined and continuous on $\mathbb{R}^+ \setminus \{0\}$.
2. Show that there exists a unique $\lambda > 0$ such that $\lim_{x \rightarrow \infty} f(x) = \lambda$.
3. Show that f is differentiable on $\mathbb{R}_*^+$ and compute $f'(x)$.
4. Sketch the graph of f .

Answer Area

1. Domain and continuity:

For $x > 0$, each $f_n(x)$ is positive and well-defined. Compare with a convergent series:

$$0 < f_n(x) = \frac{x}{n(1+n^2x)} < \frac{x}{n \cdot n^2x} = \frac{1}{n^3}$$

Since $\sum \frac{1}{n^3}$ converges, by the Weierstrass M-test, $\sum f_n(x)$ converges uniformly

on any interval $[\delta, \infty)$ with $\delta > 0$. Each $f_n(x)$ is continuous on $(0, \infty)$, so the uniform convergence implies $f(x)$ is continuous on $(0, \infty)$.

2. Limit as $x \rightarrow \infty$:

For fixed n , $\lim_{x \rightarrow \infty} f_n(x) = \lim_{x \rightarrow \infty} \frac{x}{n(1+n^2x)} = \frac{1}{n^3}$.

Interchanging limit and summation (justified by uniform convergence on $[M, \infty)$ for large M):

$$\lim_{x \rightarrow \infty} f(x) = \sum_{n=1}^{\infty} \lim_{x \rightarrow \infty} f_n(x) = \sum_{n=1}^{\infty} \frac{1}{n^3} = \zeta(3)$$

where $\zeta(3)$ is Apéry's constant.

3. Differentiability:

For $x > 0$, compute the derivative:

$$f'_n(x) = \frac{d}{dx} \left(\frac{x}{n(1+n^2x)} \right) = \frac{1}{n(1+n^2x)} - \frac{n^2x}{n(1+n^2x)^2} = \frac{1}{n(1+n^2x)^2}$$

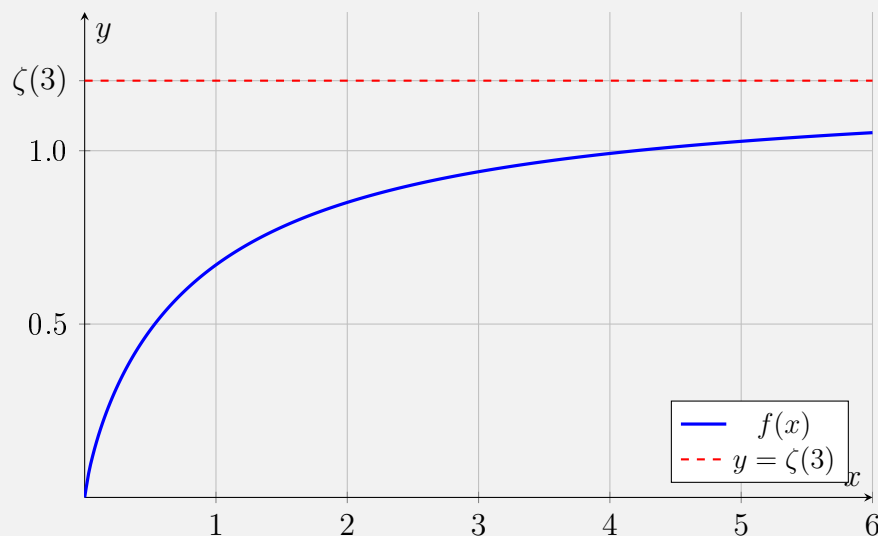
Now check uniform convergence of $\sum f'_n(x)$ on $[\delta, \infty)$ with $\delta > 0$:

$$|f'_n(x)| = \frac{1}{n(1+n^2x)^2} \leq \frac{1}{n(1+n^2\delta)^2} \leq \frac{1}{n \cdot n^4\delta^2} = \frac{1}{\delta^2 n^5}$$

Since $\sum \frac{1}{n^5}$ converges, $\sum f'_n(x)$ converges uniformly on $[\delta, \infty)$. By the theorem on differentiation of series, f is differentiable on $(0, \infty)$ and:

$$f'(x) = \sum_{n=1}^{\infty} f'_n(x) = \sum_{n=1}^{\infty} \frac{1}{n(1+n^2x)^2}$$

4. Graph of f :



Exercise 3:

Define the sequence (u_n) by $u_0 = u_1 = 1$ and for $n \in \mathbb{N}$:

$$u_{n+2} = u_{n+1} + 2u_n + (-1)^n.$$

Let $f(x) = \sum_{n=0}^{\infty} u_n x^n$ when it exists, and let R be its radius of convergence.

1. Show that $\forall n \in \mathbb{N}$,

$$1 \leq u_n \leq 2^{n+1} - 1$$

and deduce that $R \geq \frac{1}{2}$.

2. Show that for $|x| < \min(1, R)$, $f(x)$ satisfies:

$$f(x) = \frac{x^2 + x + 1}{(x+1)^2(1-2x)}.$$

3. Give an explicit expression for u_n as a function of n .

Answer Area

Given $u_0 = u_1 = 1$ and $u_{n+2} = u_{n+1} + 2u_n + (-1)^n$, with $f(x) = \sum_{n=0}^{\infty} u_n x^n$.

1. Bounds on u_n and radius of convergence:

First, prove by induction that $1 \leq u_n \leq 2^{n+1} - 1$.

Base case: For $n = 0$: $1 \leq u_0 = 1 \leq 2^1 - 1 = 1$ For $n = 1$: $1 \leq u_1 = 1 \leq 2^2 - 1 = 3$

Inductive step: Assume $1 \leq u_k \leq 2^{k+1} - 1$ for $k = n, n+1$.

Then:

$$u_{n+2} = u_{n+1} + 2u_n + (-1)^n \geq 1 + 2 \cdot 1 - 1 = 2 \geq 1$$

(using $(-1)^n \geq -1$)

And:

$$u_{n+2} \leq (2^{n+2} - 1) + 2(2^{n+1} - 1) + 1 = 2^{n+2} - 1 + 2^{n+2} - 2 + 1 = 2^{n+3} - 2$$

But $2^{n+3} - 2 \leq 2^{n+3} - 1$, so the upper bound holds.

From $u_n \leq 2^{n+1} - 1 < 2^{n+1}$, we have $\sqrt[n]{u_n} < \sqrt[n]{2^{n+1}} = 2 \cdot 2^{1/n} \rightarrow 2$.

By the Cauchy-Hadamard theorem, $R = 1/\limsup \sqrt[n]{u_n} \geq 1/2$.

2. Functional equation for $f(x)$:

For $|x| < \min(1, R)$, multiply the recurrence by x^{n+2} and sum from $n = 0$ to ∞ :

$$\sum_{n=0}^{\infty} u_{n+2} x^{n+2} = \sum_{n=0}^{\infty} u_{n+1} x^{n+2} + 2 \sum_{n=0}^{\infty} u_n x^{n+2} + \sum_{n=0}^{\infty} (-1)^n x^{n+2}$$

Rewrite in terms of $f(x)$:

$$\sum_{n=0}^{\infty} u_{n+2}x^{n+2} = f(x) - u_0 - u_1x = f(x) - 1 - x$$

$$\sum_{n=0}^{\infty} (-1)^n x^{n+2} = x^2 \sum_{n=0}^{\infty} (-x)^n = \frac{x^2}{1+x}$$

Substitute into the recurrence:

$$f(x) - 1 - x = x(f(x) - 1) + 2x^2 f(x) + \frac{x^2}{1+x}$$

Solve for $f(x)$:

$$f(x) - 1 - x = xf(x) - x + 2x^2 f(x) + \frac{x^2}{1+x}$$

$$f(x) = \frac{1 - x + \frac{x^2}{1+x}}{1 - x - 2x^2}$$

Simplify numerator and denominator:

$$1 - x + \frac{x^2}{1+x} = \frac{(1-x)(1+x) + x^2}{1+x} = \frac{1 - x^2 + x^2}{1+x} = \frac{1}{1+x}$$

And $1 - x - 2x^2 = (1+x)(1-2x)$.

Therefore:

$$f(x) = \frac{\frac{1}{1+x}}{(1+x)(1-2x)} = \frac{1}{(1+x)^2(1-2x)}$$

But wait, the problem statement says $f(x) = \frac{x^2+x+1}{(x+1)^2(1-2x)}$. Let me check the numerator again:

Actually: $1 - x + \frac{x^2}{1+x} = \frac{(1-x)(1+x)+x^2}{1+x} = \frac{1-x^2+x^2}{1+x} = \frac{1}{1+x}$. But $u_0 = 1, u_1 = 1$, so $f(x) = 1 + x + \sum_{n=2}^{\infty} u_n x^n$.

I should have: $f(x) - 1 - x = \sum_{n=2}^{\infty} u_n x^n = \sum_{n=0}^{\infty} u_{n+2} x^{n+2}$.

So the equation becomes:

$$f(x) - 1 - x = x(f(x) - 1) + 2x^2 f(x) + \frac{x^2}{1+x}$$

Solving:

$$f(x) - xf(x) - 2x^2 f(x) = 1 + x - x + \frac{x^2}{1+x}$$

$$f(x) = \frac{1 + x + x^2}{(1+x)(1-x-2x^2)}$$

Since $1 - x - 2x^2 = (1+x)(1-2x)$, we get:

$$f(x) = \frac{x^2 + x + 1}{(x+1)^2(1-2x)}$$

which matches the problem statement.

3. Explicit formula for u_n :

Perform partial fraction decomposition:

$$\frac{x^2 + x + 1}{(x+1)^2(1-2x)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{1-2x}$$

Multiply both sides by $(x+1)^2(1-2x)$:

$$x^2 + x + 1 = A(x+1)(1-2x) + B(1-2x) + C(x+1)^2$$

Solve for coefficients:

- At $x = -1$: $1 - 1 + 1 = B(1 + 2) \Rightarrow 1 = 3B \Rightarrow B = \frac{1}{3}$
- At $x = \frac{1}{2}$: $\frac{1}{4} + \frac{1}{2} + 1 = C \left(\frac{3}{2}\right)^2 \Rightarrow \frac{7}{4} = \frac{9}{4}C \Rightarrow C = \frac{7}{9}$
- At $x = 0$: $1 = A(1)(1) + B(1) + C(1) \Rightarrow 1 = A + \frac{1}{3} + \frac{7}{9} \Rightarrow A = 1 - \frac{1}{3} - \frac{7}{9} = -\frac{1}{9}$

So:

$$f(x) = -\frac{1}{9} \cdot \frac{1}{1+x} + \frac{1}{3} \cdot \frac{1}{(1+x)^2} + \frac{7}{9} \cdot \frac{1}{1-2x}$$

Expand as power series:

$$\begin{aligned} \frac{1}{1+x} &= \sum_{n=0}^{\infty} (-1)^n x^n \\ \frac{1}{(1+x)^2} &= \sum_{n=0}^{\infty} (-1)^n (n+1) x^n \quad (\text{derivative of geometric series}) \\ \frac{1}{1-2x} &= \sum_{n=0}^{\infty} 2^n x^n \end{aligned}$$

Therefore:

$$\begin{aligned} f(x) &= -\frac{1}{9} \sum_{n=0}^{\infty} (-1)^n x^n + \frac{1}{3} \sum_{n=0}^{\infty} (-1)^n (n+1) x^n + \frac{7}{9} \sum_{n=0}^{\infty} 2^n x^n \\ &= \sum_{n=0}^{\infty} \left[-\frac{1}{9} (-1)^n + \frac{1}{3} (-1)^n (n+1) + \frac{7}{9} 2^n \right] x^n \end{aligned}$$

So the explicit formula is:

$$u_n = \frac{7}{9} \cdot 2^n + (-1)^n \left(\frac{n+1}{3} - \frac{1}{9} \right) = \frac{7}{9} \cdot 2^n + (-1)^n \left(\frac{3n+2}{9} \right)$$

Or equivalently:

$$u_n = \frac{1}{9} [7 \cdot 2^n + (-1)^n (3n+2)]$$